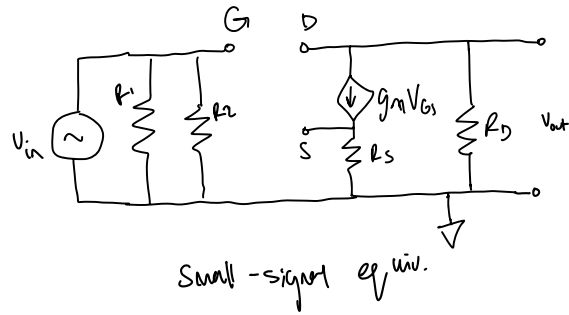
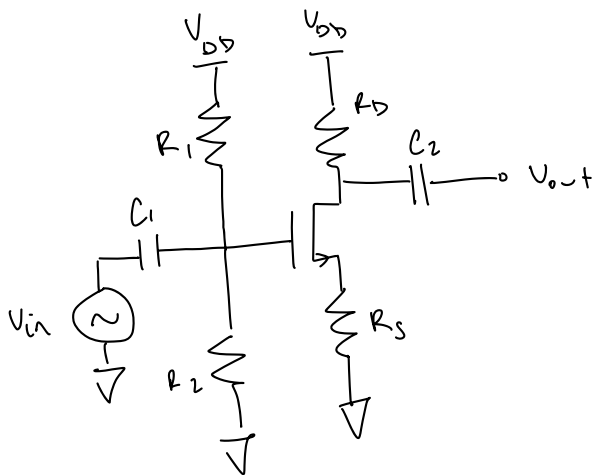




2N7000 characteristics

$$V_t = 0.8 - 3.0 \text{ V}$$

$$g_m = 100 \text{ mS} \quad (V_{DS} = 10 \text{ V}, I_D = 200 \text{ mA})$$

$$= \sqrt{2k_n I_D}$$

$$k_n = \frac{g_m^2}{2I_D}$$

$$= 25 \text{ mA/V}^2$$

Setting bias point

$$V_{DD} = 10 \text{ V}$$

$$I_D = 2 \text{ mA}$$

$$g_m = 10 \frac{\text{mA}}{\text{V}}$$

$$I_D = \frac{1}{2} k_n V_{ov}^2$$

$$V_{ov} = 0.4 \text{ V}$$

Choosing \$R_D\$ and \$R_S\$

$$A_v = \frac{-g_m R_D}{1 + g_m R_S}$$

$$\text{we want } |A_v| \geq 10$$

$$R_D \geq 1 \text{ k}\Omega + 10 R_S$$

$$R_D = 2 \text{ k}\Omega$$

$$R_S = 100 \Omega$$

constraints met!



$$\text{this gives us } V_D = 6 \text{ V}$$

$$V_S = 0.2 \text{ V}$$

$V_{DS} > V_{ov}$
safely in saturation!

$$\text{for } V_{ov} = 0.4 \text{ V, estimating } V_t = 1.5 \text{ V,}$$

$$V_G = 2.1 \text{ V}$$

Coupling capacitors

keeping it simple,

$$C_1 = C_2 = 0.1 \mu\text{F}$$

Expected values

$$A_v = \frac{-g_m R_D}{1 + g_m R_S} = 10$$

$$R_{in} = R_1 \parallel R_2$$

$$= 796 \text{ k}\Omega$$

$$R_{out} \approx R_D$$

$$\approx 2 \text{ k}\Omega$$

$$\text{then, } \frac{R_2}{R_1 + R_2} (10 \text{ V}) = 2.1 \text{ V}$$

$$R_1 = 3.9 \text{ M}\Omega$$

$$R_2 = 1 \text{ M}\Omega$$

